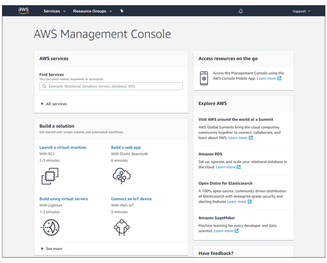
Virtualization

10.2.1

Cloud Computing and Virtualization

In the previous topic, you learned about cloud services and cloud models. This topic will explain virtualization. The terms "cloud computing" and "virtualization" are often used interchangeably; however, they mean different things. Virtualization is the foundation of cloud computing. Without it, cloud computing, as it is most widely implemented, would not be possible.

Virtualization separates the operating system (OS) from the hardware. Various providers offer virtual cloud services that can dynamically provision servers as required. For example, Amazon Web Services (AWS) provides a simple way for customers to dynamically provision the computer resources they need. These virtualized instances of servers are created on demand. As shown in the figure, the network administrator can deploy a variety of services from the AWS Management Console including virtual machines, web applications, virtual servers, and connections to IoT devices.



10.2.2

Dedicated Servers

To fully appreciate virtualization, it is first necessary to understand some of the history of server technology. Historically, enterprise servers consisted of a server OS, such as Windows Server or Linux Server, installed on specific hardware, as shown in the figure. All of a server's RAM, processing power, and hard drive space were dedicated to the service provided (e.g., Web, email services, etc.).

Dedicated Servers

The major problem with this configuration is that when a component fails, the service that is provided by this server becomes unavailable. This is known as a single point of failure. Another problem was that dedicated servers were underused. Dedicated servers often sat idle for long periods of time, waiting until there was a need to deliver the specific service they provide. These servers wasted energy and took up more space than was warranted by the amount of service provided. This is known as server sprawl.

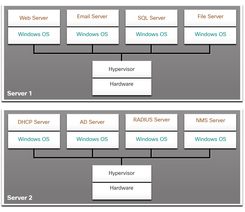
10.2.3

Server Virtualization

Server virtualization takes advantage of idle resources and consolidates the number of required servers. This also allows for multiple operating systems to exist on a single hardware platform.

For example, in the figure, the previous eight dedicated servers have been consolidated into two servers using hypervisors to support multiple virtual instances of the operating systems.

Hypervisor OS Installation



The use of virtualization normally includes redundancy to protect from a single point of failure. Redundancy can be implemented in different ways. If the hypervisor fails, the VM can be restarted on another hypervisor. Also, the same VM can run on two hypervisors concurrently, copying the RAM and CPU instructions between them. If one hypervisor fails, the VM continues running on the other hypervisor. The services running on the VMs are also virtual and can be dynamically installed or uninstalled, as needed.

The hypervisor is a program, firmware, or hardware that adds an abstraction layer on top of the physical hardware. The abstraction layer is used to create virtual machines which have access to all the hardware of the physical machine such as CPUs, memory, disk controllers, and NICs. Each of these virtual machines runs a complete and separate operating system. With virtualization, enterprises can now consolidate the number of servers they require. For example, it is not uncommon for 100 physical servers to be consolidated as virtual machines on top of 10 physical servers that are using hypervisors.

10.2.4

Advantages of Virtualization

One major advantage of virtualization is overall reduced cost:

- **Less equipment is required** – Virtualization enables server consolidation, which requires fewer physical servers, fewer networking devices, and less supporting infrastructure. It also means lower maintenance costs.
- **Less energy is consumed** – Consolidating servers lowers the monthly power and cooling costs. Reduced consumption helps enterprises to achieve a smaller carbon footprint.
- **Less space is required** – Server consolidation with virtualization reduces the overall footprint of the data center. Fewer servers, network devices, and racks reduce the amount of required floor space.

These are additional benefits of virtualization:

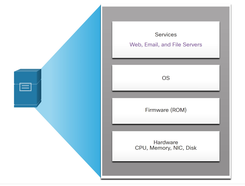
- **Easier prototyping** – Self-contained labs, operating on isolated networks, can be rapidly created for testing and prototyping network deployments. If a mistake is made, an administrator can simply revert to a previous version. The testing environments can be online, but isolated from end users. When testing is completed, the servers and systems can be deployed to end users.
- **Faster server provisioning** – Creating a virtual server is far faster than provisioning a physical server.
- **Increased server uptime** – Most server virtualization platforms now offer advanced redundant fault tolerance features, such as live migration, storage migration, high availability, and distributed resource scheduling.
- **Improved disaster recovery** – Virtualization offers advanced business continuity solutions. It provides hardware abstraction capability so that the recovery site no longer needs to have hardware that is identical to the hardware in the production environment. Most enterprise server virtualization platforms also have software that can help test and automate the failover before a disaster does happen.
- **Legacy support** – Virtualization can extend the life of OSs and applications providing more time for organizations to migrate to newer solutions.

10.2.5

Abstraction Layers

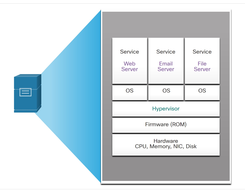
To help explain how virtualization works, it helps to use layers of abstraction in computer architectures. A computer system consists of the following abstraction layers, as illustrated in the figure:

- Services
- OS
- Firmware
- Hardware



At each of these layers of abstraction, some type of programming code is used as an interface between the layer below and the layer above. For example, the C programming language is often used to program the firmware that accesses the hardware.

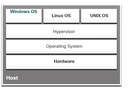
An example of virtualization is shown in the figure. A hypervisor is installed between the firmware and the OS. The hypervisor can support multiple instances of OSs.



10.2.6

Type 2 Hypervisors

A Type 2 hypervisor is software that creates and runs VM instances. The computer, on which a hypervisor is supporting one or more VMs, is a host machine. Type 2 hypervisors are also called hosted hypervisors. This is because the hypervisor is installed on top of the existing OS, such as macOS, Windows, or Linux. Then, one or more additional OS instances are installed on top of the hypervisor, as shown in the figure.



A big advantage of Type 2 hypervisors is that management console software is not required.

Type 2 hypervisors are very popular with consumers and for organizations experimenting with virtualization. Common Type 2 hypervisors include:

- Virtual PC
- VMware Workstation
- Oracle VM VirtualBox
- VMware Fusion
- Mac OS X Parallels

Many of these Type 2 hypervisors are free. However, some hypervisors offer more advanced features for a fee.

Note: It is important to make sure that the host machine is robust enough to install and run the VMs, so that it does not run out of resources.

10.2.7

Check Your Understanding – Virtualization

Check your understanding of virtualization by choosing the BEST answer to the following questions.

1. What technology separates the OS from the hardware?
   - web application
   - virtualization
   - dedicate servers
   - firmware

2. What device uses all the RAM, processing power, and hard drive space devoted to one service?
   - virtual machine
   - Type 1 hypervisor
   - dedicated server
   - virtualized server

3. Which technology is a program, firmware, or hardware that adds an abstraction layer on top of the physical hardware?
   - web application
   - virtualization
   - dedicated Server
   - hypervisor

4. What are the major advantages of virtualization? (Choose three)
   - [x] requires less equipment
   - [ ] less abstraction
   - [ ] fewer operating systems
   - [x] faster provisioning
   - [x] increased server uptime

5. Which technology is installed on top of the existing OS to create and run VM instances?
   - Type 2 Hypervisor
   - web applications
   - virtualization
   - dedicated servers

Check

Show Me

Reset